

Foundations of Data Science for Biologists

Introduction to SQL

BIO 724D

2025-NOV-18

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Why learn SQL?

SQL is present wherever you use a computer, mobile device, and much more

Personal: playlists, contacts, calendar, bookmarks, locations, photos, passwords

Shopping: order tracking, order history, favorites, suggestions, payment methods

Travel: booking tickets and hotels, tracking arrivals/departures, booking a ride

Banking: account balances, ATM transactions, loan payments, credit card statements

Healthcare: electronic health records, billing, appointments, test results

Education: registration, payroll, transcript, progress towards degree, reimbursements

Publication: manuscript tracking, bibliographic search engines, bibliography managers

Data repositories: nearly all data in nearly all fields of study

Introduction to SQL

What is SQL?

SQL (**S**tructured **Q**uery **L**anguage) is designed for interacting with tabular data
Specifically, it implements **relational** data structures



SQL is designed:

- for **powerful data queries** using a simple and compact syntax
- to enforce **data integrity** during data entry and updating
- for **highly efficient** search, sort, grouping, summarizing, joining, etc. operations
- to be **massively scalable** (billions of rows, thousands of columns)

SQL is a **domain-specific language** (DSL)

- Designed to meet a specific set of needs

- Best-in-class for its intended purpose, bad-to-awful for most other purposes

SQL is designed to work with relational data structures

observations				
seq	genus_ioc	species_ioc	obs_date	location_name
26134	Bucorvus	leadbeateri	2017-10-03	Tarangire NP
26140	Sporopipes	frontalis	2017-10-03	Tarangire NP
26149	Agapornis	personatus	2017-10-03	Burunge WMA
26155	Serinus	flavivertex	2017-10-03	Burunge WMA
26158	Pytilia	melba	2017-10-03	Burunge WMA
26161	Melierax	poliopterus	2017-10-03	Burunge WMA
26164	Halcyon	leucocephala	2017-10-04	Tarangire R Camp
26167	Pytilia	melba	2017-10-04	Tarangire R Camp
26170	Cercotrichas	leucophrys	2017-10-04	Tarangire R Camp

species					
seq	genus_ioc	species_ioc	num_sbsp	species_familiar	IUCN
79816	Carduelis	carduelis	10	European Goldfinch	LC
79821	Serinus	serinus	1	European Serin	LC
79826	Serinus	canicollis	3	Cape Canary	LC
79831	Serinus	flavivertex	3	Yellow-crowned Canary	LC
79836	Spinus	tristis	5	American Goldfinch	LC
79841	Spinus	psaltria	1	Lesser Goldfinch	LC

locations				
location_name	province	country_name	climate	geoloc
Buff Bay	Po	Jamaica	Af	18.2337,-76.6603
Bukit Timah NR	Sin	Singapore	Af	1.3492,103.7774
Bukit Tinggi	Phg	Malaysia	Af	3.4195,101.8421
Burleigh Head NP	Qld	Australia	Cfa	-28.0926,153.4581
Burunge WMA	Aru	Tanzania	Bsh	-3.7802,35.9617
Buskett Gardens	Mal	Malta	Csa	35.8584,14.4001

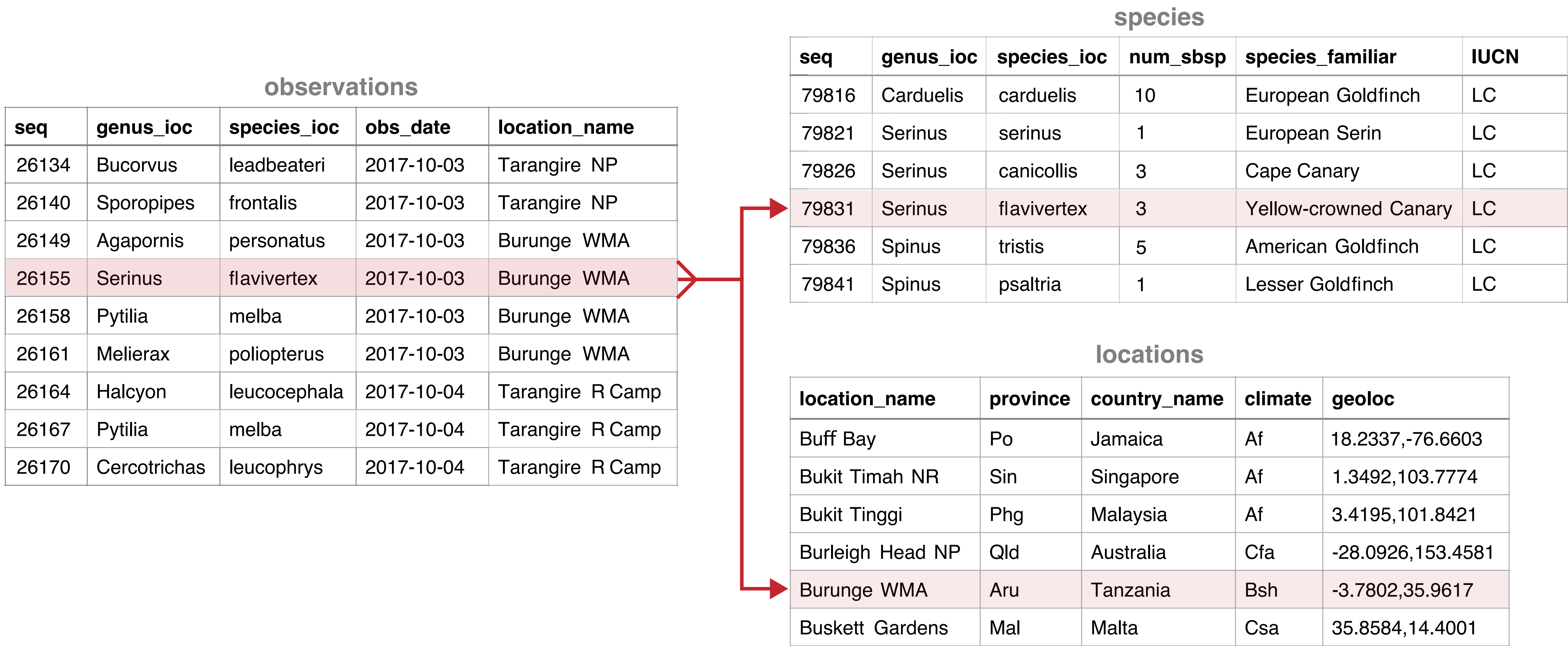
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SQL is designed to work with relational data structures



Relational design removes redundant information by spreading it across tables:

- (1) reduces errors,
- (2) simplifies updates,
- (3) saves space,
- (4) speeds up queries

Key features of relational data structures

Every table contains a **primary key** (PK) consisting of 1 column (or, occasionally, more)

- Every row of the PK column must contain a value (no blanks or NULL values)

- Every value in the PK column must be unique (no duplicate values)

Every table should contain at least one **relation**

- Relation = a column that references a column in another table (typically its PK)

- Both columns must have the same data type; typically they contain overlapping values

The order of rows in a table is not consistent or stable

- This allows for highly efficient query and sort operations

- Retrieving data in a guaranteed sort order requires an explicit definition every time

SQL is limited to a small set of operations

Only four kinds of operations are permitted with relational databases

Create, Read, Update, Delete (CRUD)

CREATE to create a new database, table, or relation; **INSERT** to add rows to a table

SELECT to query (retrieve information from) a database

UPDATE to change information in a table

DROP to remove a database, table, or relation; **DELETE** to remove rows from a table

SQL has some additional keywords, but most work with the above set

Clauses: **WHERE**, **JOIN**, **GROUP BY**, **SORT BY**, **LIMIT**, etc.

Operators: **=**, **>**, **AND**, **NOT**, **IN**, **LIKE**, **BETWEEN**, etc.

Functions: **COUNT**, **DISTINCT**, **MIN**, **MEAN**, etc.

dplyr and **Pandas** implement some SQL-like functions (and were inspired by it)

Practice database

Practice database: birding observations

Full implementation consists of 9 tables:

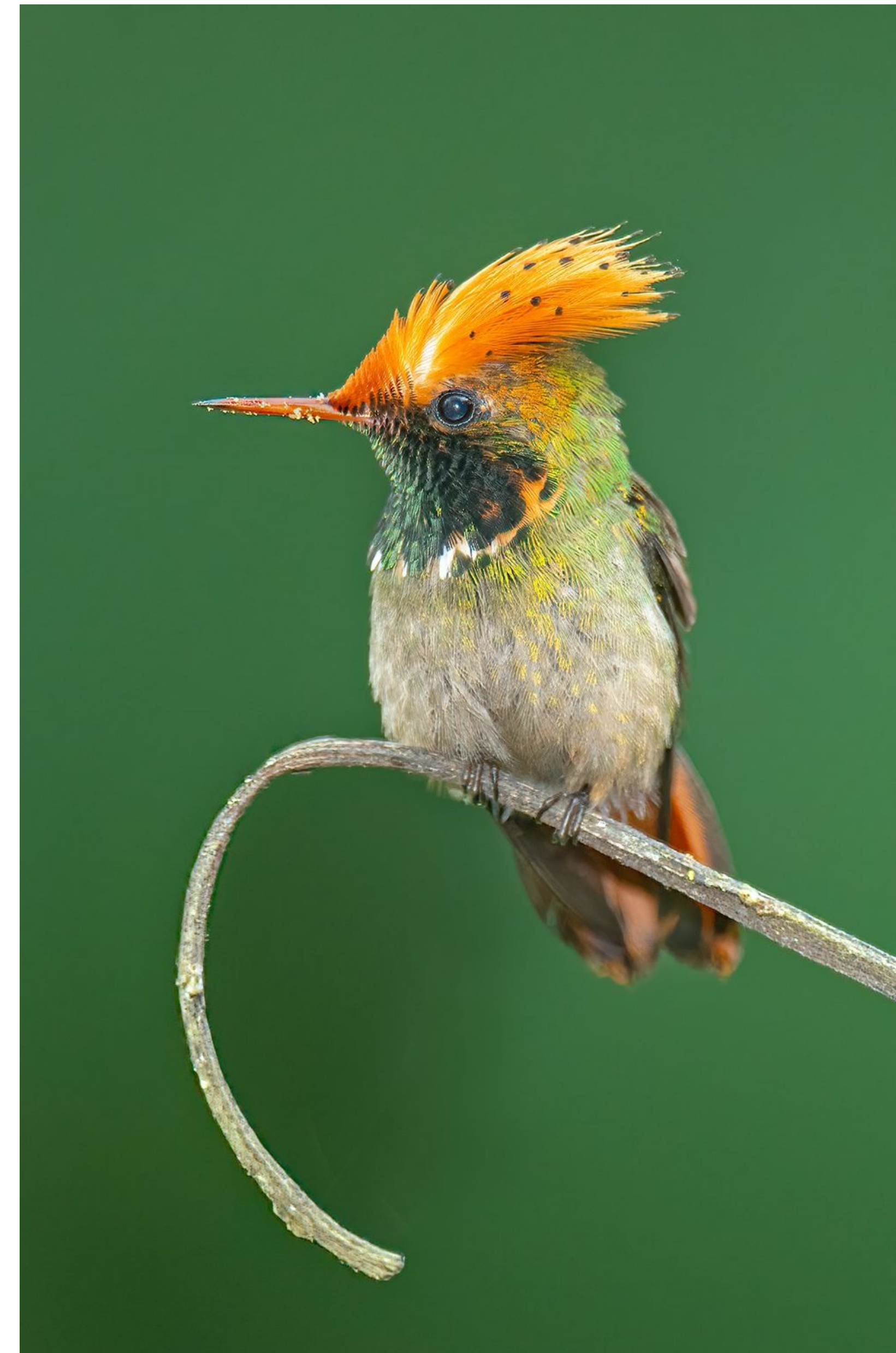
- 1 records field observations: the heart of the database
- 4 define taxonomy: species, genera, families, orders
- 3 define place: location, country, bioregion
- 1 defines context: trips/residences

Simplified version:

- Omits 2 tables, omits some columns from every table
- Includes >13.5K observations

Lophornis stictolophus
Spangled Coquette

Photo credit: Steve Gettle



Points to keep in mind about the database

Taxonomy follows the International Ornithological Congress (IOC)

- Scientific name: single word at each rank, follows rules of zoological nomenclature

- Familiar (common) name: single name, attempt to reconcile multiple synonyms

- Sequence of taxa within each rank: provides consistent ordering in lists

The **content** of the taxonomic tables differs by rank

- orders** and **families**: complete list of taxa recognized by the IOC

- genera** and **species**: only taxa that have been observed

- subspecies**: not in a separate table, but recorded in **observations** table

Primary keys are based on how the database is queried and updated

- orders**, **families**, and **genera**: taxon name; **species**: IOC sequence (or binomen)

- locations** and **trips**: brief name; **observations**: temporal sequence

Practice database: birding observations

taxonomy

main.orders

order_ioc

seq

order_familiar

taxonomy

main.families

family_ioc

seq

order_ioc

family_familiar

family_contents

niche

distribution

taxonomy

num_genera

num_genera_x

num_species

num_species_x

num_species_t

main.genera

genus_ioc

seq

family_ioc

genus_contents

taxonomy

num_species

num_species_x

num_species_t

main.species

seq

genus_ioc

species_ioc

species_familiar

species_synonyms

taxonomy

conservation

endemism

num_subspecies

num_subspecies_x

place

main.bioregions

bioregion_name

biorealm_name

definition

countries_edge

counties_entire

description

main.countries

country_name

country_formal_en

country_formal_self

country_iso

prov_count

continent_name

area_total

area_water

coastline

bioregion_list

bioregion_count

climate_count

species_total

species_resident

species_endemic

main.locations

location_name

location_specifics

enclosed_locs

province_name

country_name

bioregion_name

climate

geoloc

elevation

protection

bli_status

earliest

latest

location_type

circumstances

context

main.trips

trip_name

description

countries

trip_start

trip_end

purpose

companions

trip_notes

records

main.observations

seq

genus_ioc

species_ioc

id_confidence

id_method

subspecies_ioc

subspeices_method

individuals

date_observed

date_precision

time_observed

weather

location_name

location_granular

geoloc_coord

geoloc_prec

platform

naturalness

range

social

trip_name

companions

field_notes